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GPH 417 — Past Questions & Materials
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Course: GPH 417
Institution: Federal University of Petroleum Resources, Effurun
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Levels: 200, 400

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Paper 1 of 2 — GPH 417 - 2024/2025 - First Semester - SEISMIC EXPLORATION MTD II

Level 200 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN



DEPARTMENT OF APPLIED GEOPHYSICS

FIRST SEMESTER: B.Sc. (GEOP. & GEOLOGY) DEGREE EXAMINATION 2024/2025 SESSION

COURSE: GPH 417 (3 Units) SEISMIC EXPLORATION MTD II **TIME ALLOWED:** 2¹/₂ Hrs.

INSTRUCTION: ANSWER FOUR (4) QUESTIONS IN ALL. TWO (2) QUESTIONS FROM EACH SECTION.
(Seismic sections must be submitted along with your answer Booklet)

SECTION-A

Question 1:

(a) Briefly explain any Four (4) direct hydrocarbon indicators observed in seismic section. (12½ marks)

(b) If the velocity of gas sand is 1900m/s and that of water sand is 2500m/s and the saturated zone is 30m thick. How long will it take a reflection from an interface to reach the surface through the gas zone? (5 marks)

Question 2:

(a) Why is it easier to locate hydrocarbon structurally entrapped than in stratigraphic traps? (3 marks)

(b) Write short notes on;

(i) Limestone reefs

(ii) Facies changes (10 marks)

(c) What is the P-wave velocity of sandstone containing 70% quartz, 10% calcite and 20% water filled porosity? Given that P-wave velocity of quartz = 5200m/s, water = 1500m/s and calcite = 3500m/s. (4½ marks)

Question 3:

(a) Using a simple ray diagram of reflection from a horizontal interface, briefly explain with equations

(i) Normal moveout

(ii) Critical distance

(b) Using sketches, distinguish between 2-D and 3-D seismic surveying. Which one will you recommend for petroleum exploration? Justify your answer. (9 marks)

SECTION-B

Question 4:

(a) Attempt a brief classification of the various stratigraphic and structural features identified on seismic data. (2½ marks)

(b) Write short notes on any five (5) fluid flow features found on seismic section. (10 marks)

(c) With the aid of diagram(s), discuss Vertical Seismic Profiling and state its applications. (5 marks)

Question 5:

(a) Briefly discuss the characteristics of seismic events (6 marks)

(b) Reflection events that occur on a seismic trace can be divided into several types, discussed the major types known to you? (8 marks)

(c) With the aid of diagram(s) and example(s) briefly discuss the term seismic reflection terminations. (3½ marks)

Question 6:

(a) What are the primary indications of Faulting in a seismic section? (2 marks)

(b) From the seismic section in figure-1 below, answer the following questions:

(i) Identify and mark three key horizons, H1, H2 and H3 based on their reflection configuration. (1½ marks)

- (ii) The marked horizons will divide the seismic section into segments, top-H1, H1-H2, H2-H3 and H3-bottom. Describe these four segments of the seismic section in terms of reflection strength, reflection character/configuration and reflection continuity. (6 marks)
- (iii) Identify and clearly mark/highlight the **major faults** observed in the seismic section using a pen or pencil, indicate the **fault type** and mark the direction of the downthrown side of the fault. (2 marks)

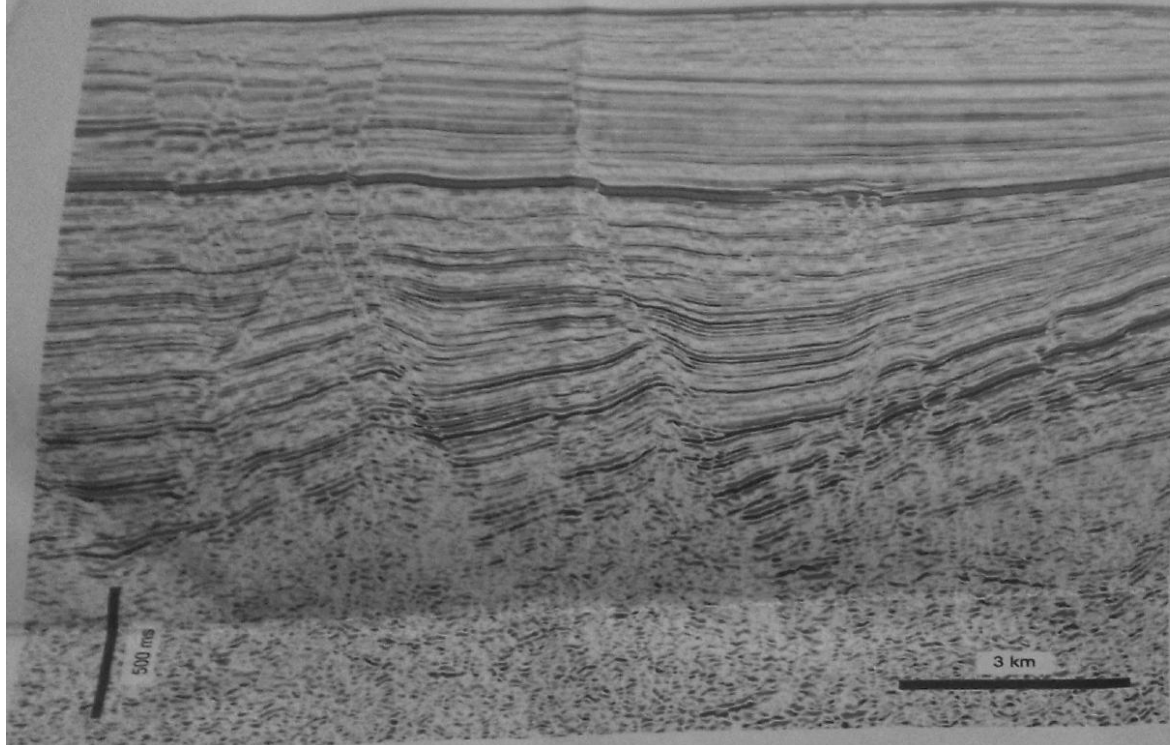


Figure-1: 2D Seismic Section Courtesy of Fugro MCS and the Virtual Atlas

- (i) Define and state the equation of the root mean square velocity (V_{rms}) (2½ marks)
- (ii) The table shows a sample of a checkshot data recorded at intervals of 50ms TWT. Compute the interval velocities and complete the table below.

MD (m)	TWT (ms)	Interval Velocity- V_{int} (m/ms)
0	0	
107.16719	50.00000	
145.88991	100.00000	
185.51532	150.00000	
226.05066	200.00000	
267.49636	250.00000	
309.85216	300.00000	

(3½ marks)

Print the attached figure in coloured.

Paper 2 of 2 — GPH 417

Level 400 • First Semester • Handout

GPH 417 SEISMIC EXPLORATION METHOD II (3 CREDITS)

Seismic response of various stratigraphic and structural features.

Different features identified on seismic data, exhibit varying seismic responses which may be attributed to subsurface lithological variations, fluid content, fluid properties, sediment deposition, diagenesis/post-depositional processes and tectonic/structural evolution of the basin under study. These features could either be Stratigraphic or Structural features. Here we break them down into two main groups:

- Fluid flow features
- Seal-by pass systems

1.0 Fluid flow Features

Fluid flow features represent anomalies generated during past and present subsurface flow of fluids (oil, gas, brine, groundwater and magmatic fluids) from source to the seabed or land surface (e.g. Seismically observed fluid flow features are typically represented by amplitude anomalies emanating

from a wide range of geological structures such as pockmarks, mud volcanoes, gas hydrates, chimneys, pipes, sediment injections, carbonate mounds, seeps and related diagenetic phenomena. The type of structure generated depends on a variety of parameters as for instance the

source of fluid, the flow type, the structural setting and the nature of the hosting sediment for the fluid flow. Fluid flow features are of major importance when analyzing basins with respect to the presence and extent of hydrocarbons and other fluids. Traditionally, fluid flow features are used as indicators of fluid venting to the seafloor or land surface (e.g. seabed pockmarks) or as indicators of overpressure build up in remobilized source intervals (e.g. sandstone intrusions).

Since the first observation and recognition of focused fluid flow features was reported, the interest has been increasing and in the past decades amplified rapidly due to the growing quality of 3D seismic

data and the realization of the impact fluid flow features may have on hydrocarbon plumbing systems particularly concerning reservoir connectivity, migration and risk assessment. Apart from impacting hydrocarbon plumbing systems the occurrence of fluid flow features may also have implications for the extraction of groundwater and geothermal energy, storage of CO₂ and nuclear waste, and for the global organic carbon cycle of the Earth.

They may also be grouped according to the cause of their formation. Such a division comprises three

major groups: a) subsurface sediment remobilization, b) vertically focused fluid flow, and c) laterally extensive fluid flow. An overview of the different types of fluid flow features included in each of the three groups is given in Fig. 1 & 3 below, and the section below briefly presents the three

groups. Note that some fluid flow features relate to more groups e.g. mud volcanoes which are formed due to subsurface sediment remobilization and vertically focused fluid flow.

Fig 1

1.0.1 Focused fluid flow and subsurface sediment remobilization

Subsurface sediment remobilization generally involves fluid flow as a cause of overpressure build-up

facilitating remobilization. Remobilization of clastic sediments may generate sand intrusion complexes

and mud volcano systems (Fig. 1), which represent some of the most thoroughly described fluid flow

features exemplified in many case studies.

a) Sand injection complexes (Sandstone intrusions or Sand Injectites)

The remobilization of clastic sediment is ascribed to fluidization of either an overpressured sand source

unit upon seal failure and overpressure build-up. Sand injection involves the forced emplacement of sand

in an adjacent lithology, and this occurs when fluidized sand is forcibly injected into a host stratum, resulting

to the formation of networks of intrusions (Fig. 1, 2 & 3). These networks of intrusions are suggested to

form in response to the failure of a low permeable sealing lithology caused by hydraulic fracturing, which

results to either upward and/or lateral escape of fluidized sand.

In seismic data the intrusions are recognized as high-amplitude reflections cross-cutting the surrounding strata giving rise to a variety of geometries including dykes, sills, conical, saucer-shaped

and wing-like intrusions.

Fig 2

1.0.2 Vertically focused fluid flow

During mainly vertical and highly focused fluid flow pockmarks, chimneys, pipes, and other discrete seabed features may be formed.

a) Pockmarks

The seafloor expression of vertical focused fluid flow can be manifested in pockmarks forming as a result of expulsive fluid venting. Pockmarks are shallow seabed depressions a few metres to tens of metres deep. The diameter is typically several tens of metres, but may vary from less than 5 m to several hundreds of metres. Pockmarks are generally formed in soft, fine-grained seabed sediments by

the escape of fluid or gas into the water column (Hovland and Judd, 1988). Pockmarks may be circular,

elliptical or composite when individual pockmarks merge into each other. They may occur in strings (e.g., along faults). On seismic sections, pockmarks are expressed as minor depressions of the seafloor

(Fig. 1 & 3). Pockmarks are often located above vertical zones of degraded seismic signal indicating hydrocarbon leakage and/or above high amplitude anomalies that may represent shallow gas reservoirs.

A dip-azimuth map at the seabed off the Nigerian continental shelf. Mud volcanoes (large cones) and numerous pockmarks are present (Loseth et al., 2009)

b) Chimneys and pipes

Chimneys and pipes form during vertically focused fluid flow and are recognized as vertical extensive

either low- or high-amplitude anomalies on seismic data (Fig. 1 & 3). Seismic chimney is defined as a

vertical zone (wide or narrow) of low-amplitude and distorted seismic reflections, while a pipe is defined as a narrow (< 300 m) vertical zone consisting mainly of stacked high-amplitude anomalies.

Gas chimneys are normally associated with dimmed and distorted amplitudes, while pipes may consist

of vertically stacked either high- or low amplitude anomalies. Chimneys and pipes are generated by several mechanisms related to fluid flow such as free gas migration, fracturing, and cementation.

1.0.3 Laterally extensive fluid flow features

Laterally extensive fluid flow features may arise whenever fluid flow is focused in certain stratigraphic

levels, as for instance high porosity beds, or at specific subsurface depths and include structures such

as gas hydrates, free gas zones, bottom simulating reflections (BSRs), polygonal faults, hydrocarbon-

related diagenetic zones (HRDZs) and diagenetic fronts (Fig. 1 & 3). These types of structures are often

recognized as high-amplitude anomalies in the seismic data.

a) Gas Hydrate

The presence of gas hydrates (ice-like crystals) and associated free gas zones and BSRs depend on specific conditions of temperature and pressure (Fig. 1 & 3). Gas hydrates (or clathrates) are crystalline, ice-like compounds composed of water and natural gas (most commonly methane) (Hovland and Judd, 1988). Their appearance is dependent on the supply of gas, temperature and pressure. They are normally found in relatively deep water and are located in areas with active seepage. Bottom simulation reflectors (BSRs) are seismic reflections that mimic the seabed shape and

are located in the shallow part of seismic sections. BSR is a common seismic expression of gas hydrates. Gas hydrates are important features due to the potential massive release of gas to the ocean

and atmosphere which may arise if pressure and temperature conditions are changed and the hydrates dissociate and/or dissolve. Gas hydrates furthermore represent potential drilling hazards and

are normally avoided in well planning.

b) Polygonal faults

Polygonal faults typically occur in certain stratigraphic intervals (tiers) and are formed during early burial upon compaction-related dewatering and hence is associated to flow and release of fluids (Fig. 1

& 3). They form mainly in mudstones and usually have polygonal map-view geometries

Seismic expression of polygonal fault systems (Cartwright et al., 2007)

Fig 3 (Andresen 2012)

2.0 Seal-by pass systems (SBS)

We define seal bypass systems (SBS) as “large-scale (seismically resolvable) geological features embedded within sealing sequences that promote cross-stratal fluid migration and allow fluids to bypass the pore network.” Here we consider two main groups of SBS (Table 1), consisting of: (1) faults,

and (2) intrusions.

Table 1: Classification of SBS (Cartwright et al., 2007)

Examples of trap-defining faults include many of the classic tilted fault block traps of the North Sea Basin, or thrust hanging-wall traps of foreland thrust and fold belts. Another example are the listric/growth faults in the Niger Delta Basin. Examples of supratrap faults include, polygonal fault systems as widely developed in seals of Cenozoic age in northwest Europe and on the Atlantic margins.

Example of Trap defining fault: Listric faults in 3D seismic from the Niger Delta Basin (Maloney et al., 2011)